

Validation

No	Requirement	Validate	Expected	Result
VAL-01	A first-time player can complete the full gameplay loop.	Ask a new player to start, move, avoid and shoot asteroids, gain points, lose, and restart or close the game.	The player completes each task without guidance from the developer.	PASS
VAL-02	The player can understand the controls and use the ship to survive.	Ask a new player to move around and avoid incoming asteroids. Also ask them to destroy them.	The player can deliberately dodge and shoot, and explain the controls of the game.	PASS
VAL-03	The player can understand their score and compare runs.	Ask a new player to score points and compare them with the previous highscore.	The player identifies both values correctly and can tell which run scored better.	PASS
VAL-04	The player can recover from losing and start another attempt.	See if a new player can properly understand when the game is over and if they can start new run	The player understands why the run ended and starts a fresh attempt using the displayed instructions.	PASS
VAL-05	The player experiences increasing difficulty during a run.	Ask a new player to score some points so that they can experience the increase of difficulty.	The player notices the increase of asteroid speed and background moving.	PASS
VAL-06	The player can recognise and benefit from rewards.	Let the player collect a falling reward and see if they understand its benefits.	The player distinguishes the reward from asteroids and senses their effect.	PASS
VAL-07	The player can find and use the game menu.	Ask a new player to open the menu and change the difficulty of the game. See if they understands what the difference between the 2 modes is.	The player finds the menu, reads the instructions and returns to gameplay without developer guidance.	PASS

Verification

No	Requirement / test	Verify	Expected	Result
VER-01	The player must be able to control the spaceship with a keyboard.	Press W, A, S and D separately, then test the arrow keys.	Each key moves the ship in its indicated direction. Releasing the movement keys stops player-controlled movement.	PASS
VER-02	Asteroids must spawn and move from top to bottom.	Start a run and see if asteroids move all the way through top to bottom and then respawn at the top	Asteroids enter from above, travel downward and continue spawning during gameplay.	PASS

VER-03	The player must be able to fire projectiles.	Press SPACE and fire a projectile, see if you can fire another one, while there is a current one on the screen.	A projectile appears at the ship and travels in the firing direction. The player can fire again once the previous projectile is gone.	PASS
VER-04	Projectiles must destroy asteroids when they collide.	Fire at an asteroid and see if both the projectile and asteroid disappear	A hit removes the asteroid and the projectile. A clear miss does not destroy the asteroid.	PASS
VER-05	Collisions between the ship and asteroids must be detected correctly.	Hit the ship without a shield into an asteroid and see if the run ends.	An overlapping collision ends the run. A clear non-contact pass does not end it.	PASS
VER-06	The player must be able to restart after losing.	End a run and try to start a new one using R or Enter	A fresh playable run starts, the score resets to zero and objects/projectiles from the previous run do not carry over.	PASS
VER-07	The game must track and display the player's score.	Destroy asteroids and see if the score increases.	Score starts at 0, increases by 1 for an asteroid and updates on screen.	PASS
VER-08	Asteroids must become faster when the score is higher.	Start with score = 200 (change code) and see if asteroids are faster than score = 0	Crossing time is shorter at the higher score.	PASS
VER-09	The Game Over menu should show Score and Highscore.	Finish a run with score 10, and then one with score 5. See if score and highscore in game over menu show the right values.	Score matches the finished run. High Score retains the best completed run and updates when beaten.	PASS
VER-10	Temporary upgrades or rewards should appear during gameplay.	Collect a reward and see if it disappears and gives you benefits. See if another one appears after 10 seconds.	Rewards appear during gameplay. Collecting one removes it and grants its benefit once. Later rewards can appear.	PASS
VER-11	The game should contain a basic main menu.	Press M during GAME_RUNNING or GAME_OVER. See if you can exit menu with M or Enter	The menu opens, gameplay pauses while it is open, and returning resumes the same run without resetting the score.	PASS
VER-12	The game could have different game modes.	Open menu and change game mode	The selected label changes. EASY keeps the ship facing and firing upward. HARD turns the ship with movement and fires in its facing direction.	PASS
VER-13	The player can close the game at the end of the gameplay loop.	Start the game and try closing it with: Esc, Ctrl+C, Mouse. See if any crashes occur	The application exits without freezing or showing a crash.	PASS